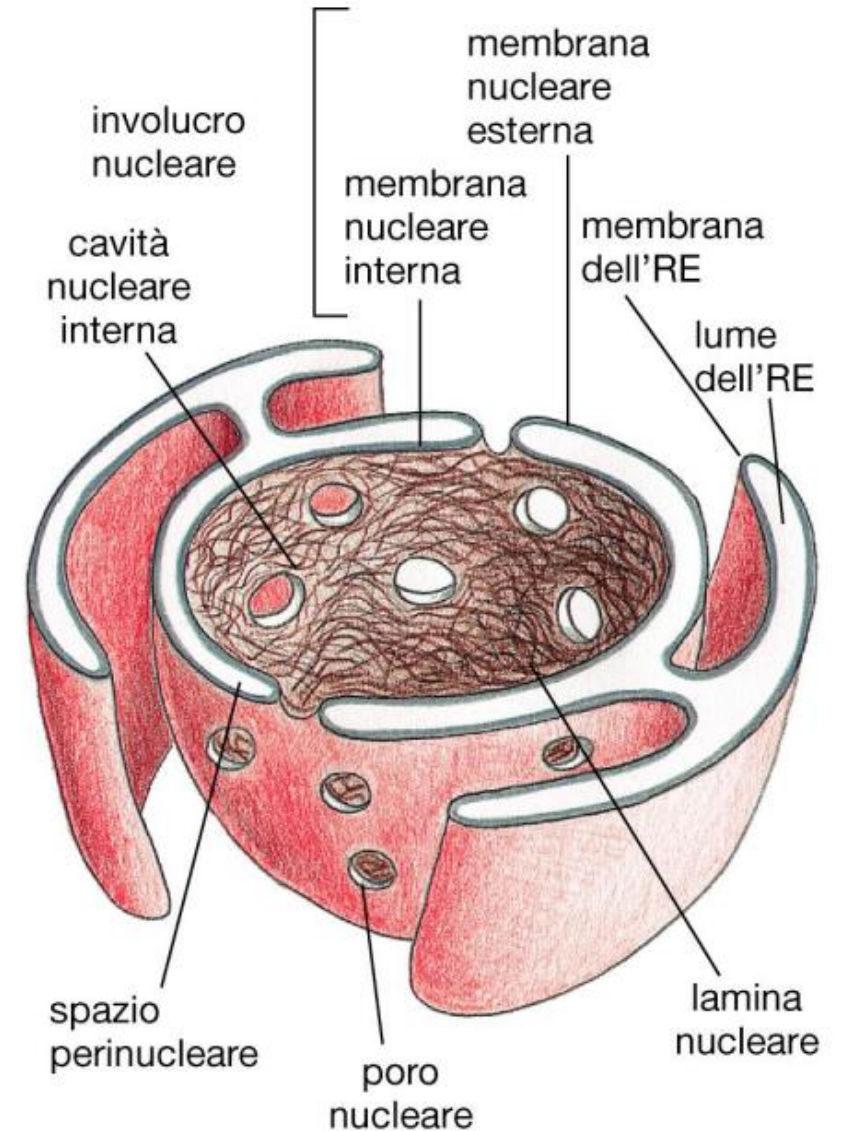
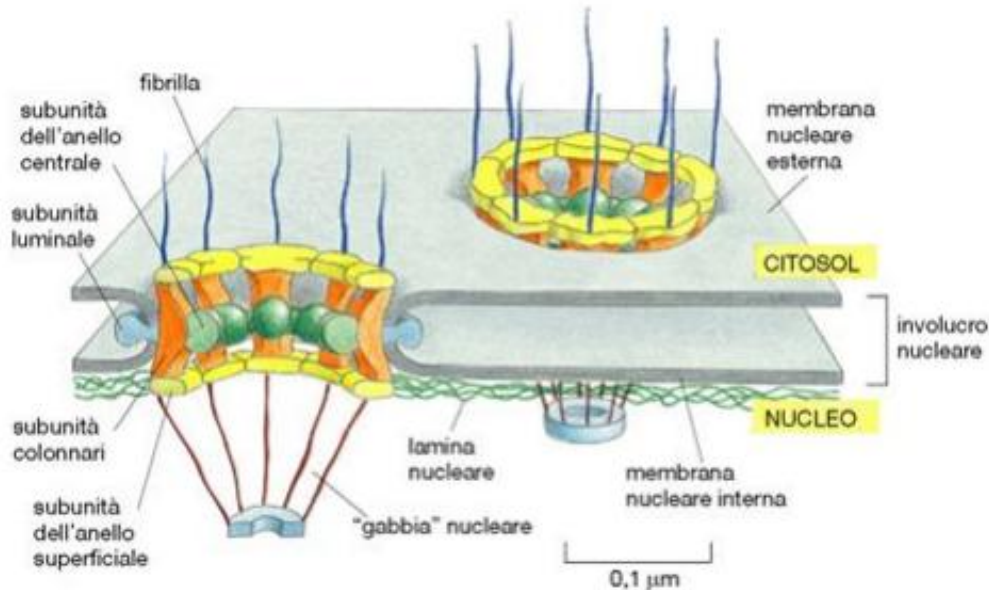


L'Envelope Nucleare

L'involucro nucleare (NE) è costituito da due membrane concentriche, la membrana nucleare interna (INM) e la membrana nucleare esterna (ONM).

Questa doppia membrana è punteggiata dai complessi dei pori nucleari (NPC).

Il lume tra i bilayer, noto come spazio perinucleare è contiguo al lume del reticolo endoplasmatico (ER).

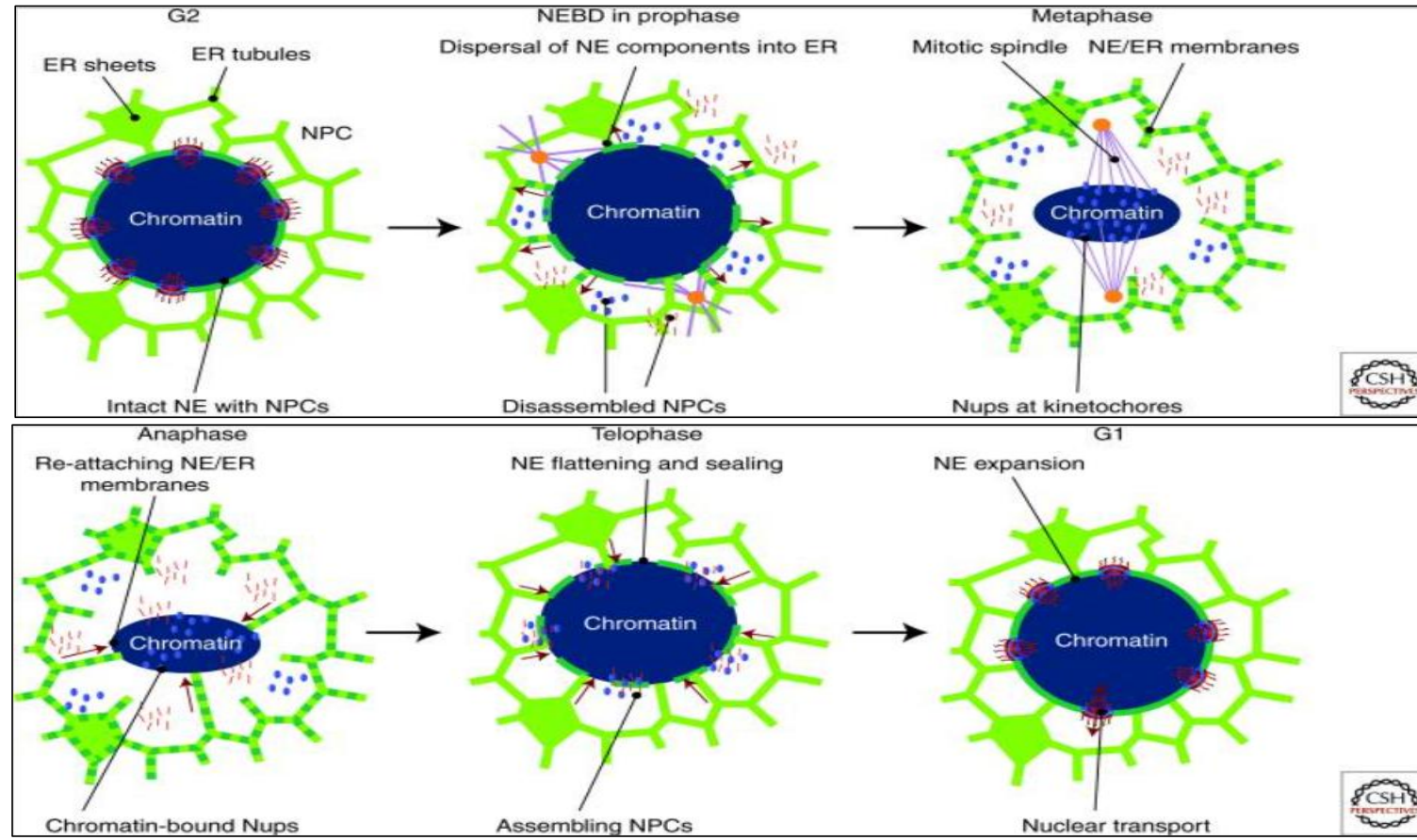
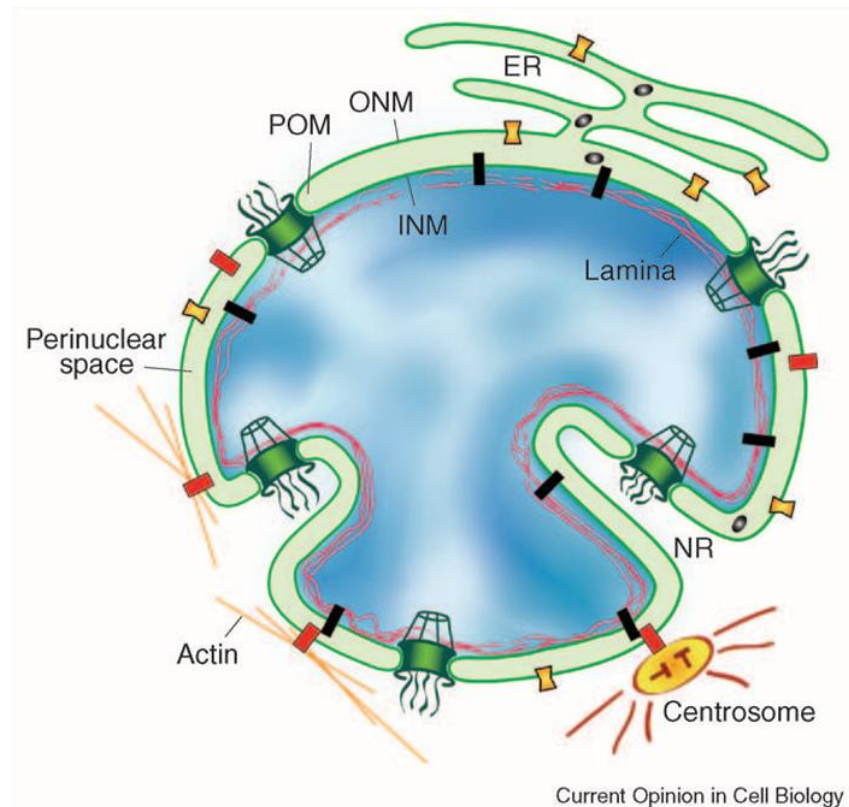


L'Envelope Nucleare

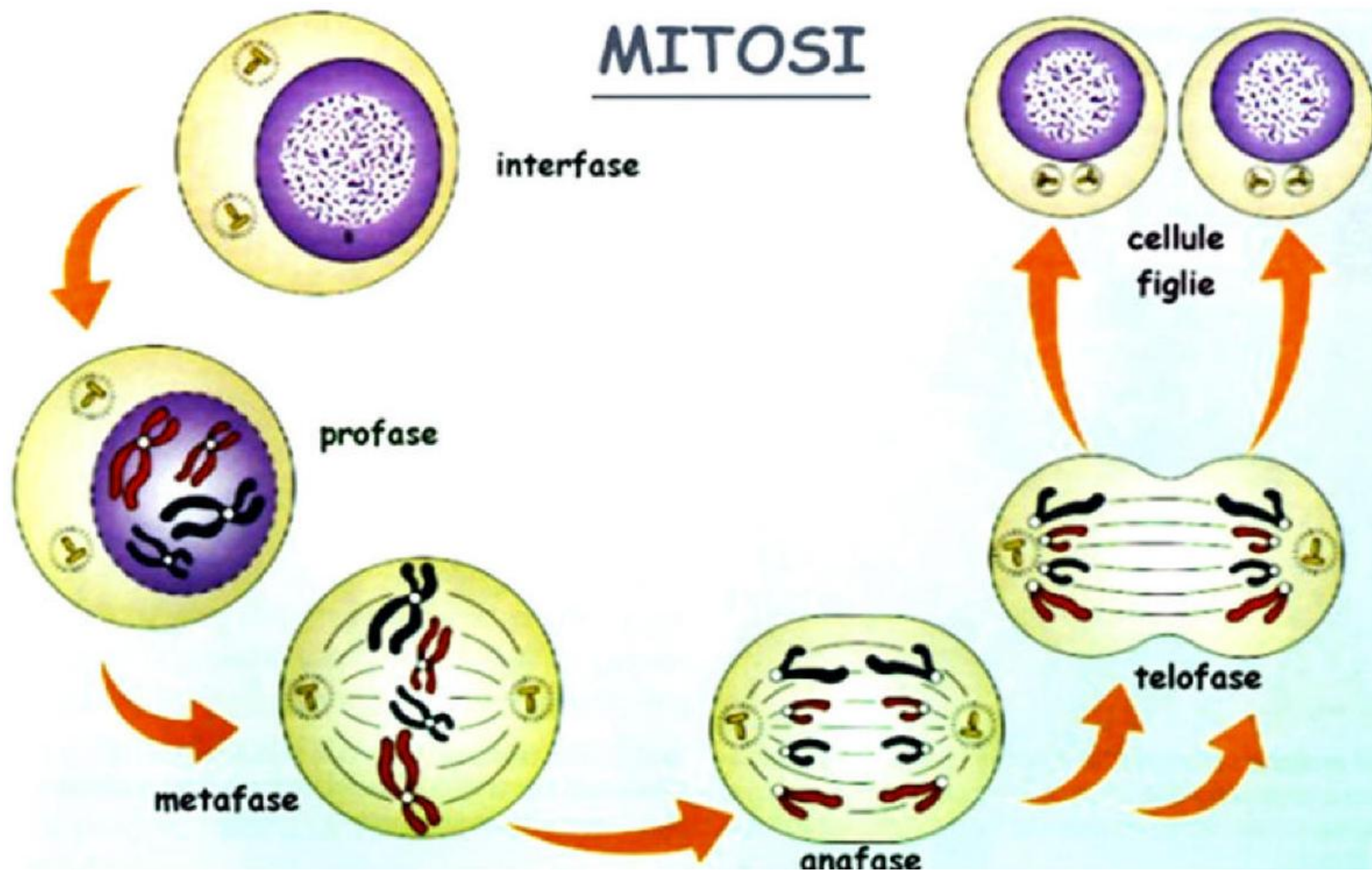
L'INM è in contatto con la lamina nucleare sottostante e con regioni di cromatina, mentre l'ONM è associata sia alla rete citoscheletrica di actina che al centrosoma.

Il compartimento perinucleare condivide molti componenti con il lume dell'ER.

Mentre gli eucarioti più primitivi subiscono una “mitosi chiusa”, in cui i microtubuli del fuso possono formarsi all'interno del nucleo o sono in grado di penetrare una membrana nucleare intatta. Al contrario, il NE delle cellule degli eucarioti superiori si disintegra completamente durante la divisione cellulare per consentire al fuso mitotico di accedere ai cromosomi. Di conseguenza, ogni cellula che si divide deve riformare il NE e ristabilire l'identità del compartimento nucleare.



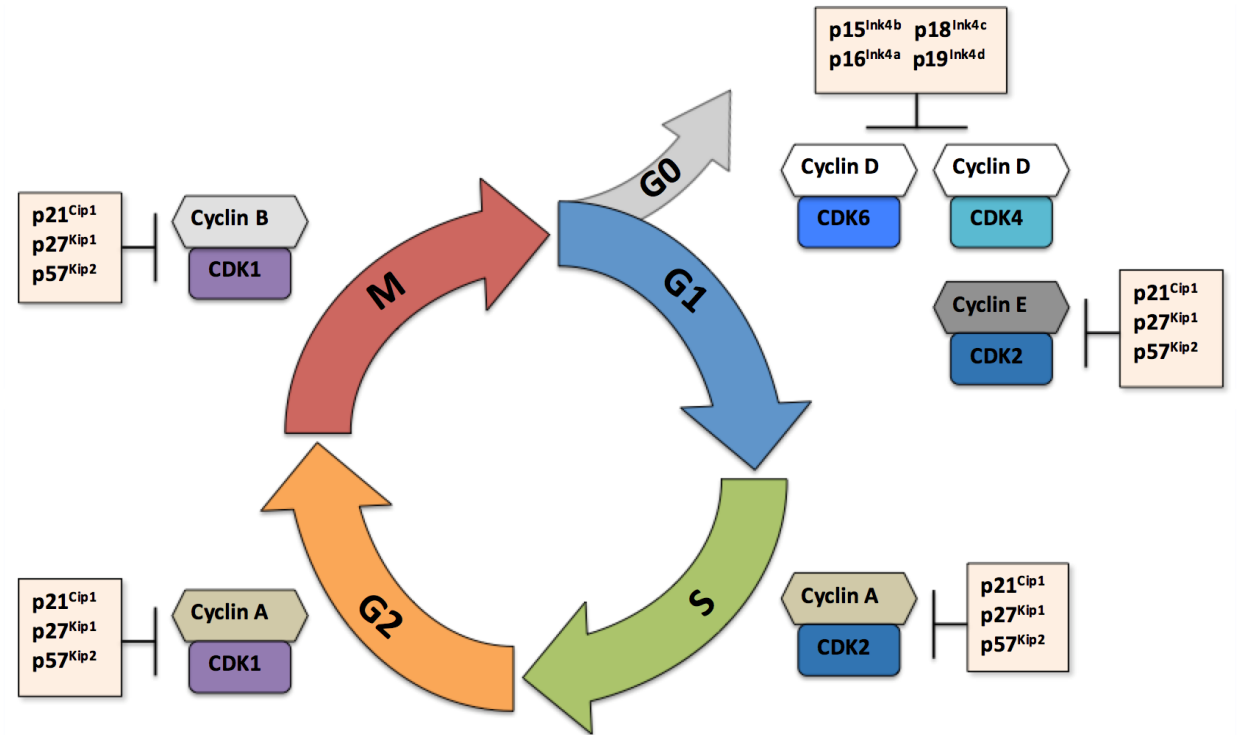
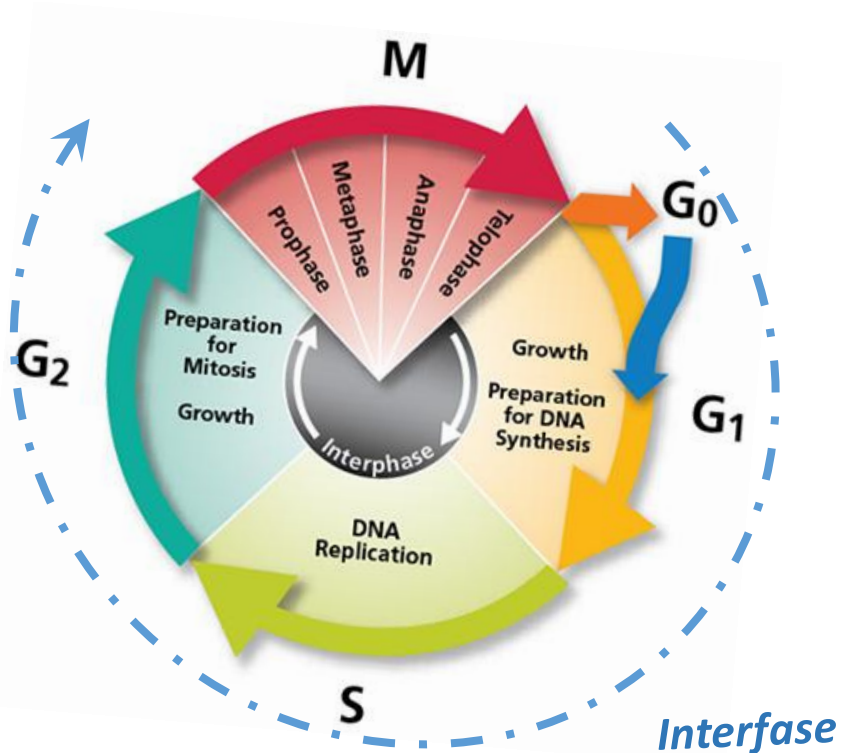
Divisione Cellulare: Mitosi e Meiosi



Divisione Cellulare: MITOSI

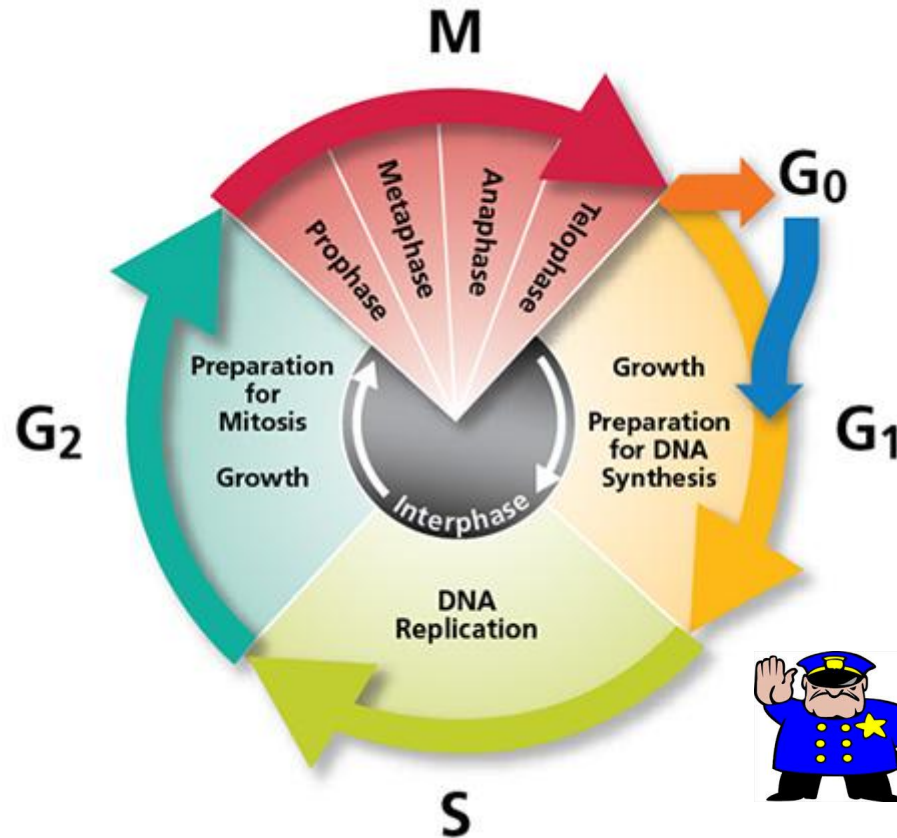
Interfase: G1-S-G2

G1: è la fase di crescita cellulare durante la quale la cellula aumenta di dimensione, espande il numero e la dimensione degli organelli, produce ed accumula le molecole necessarie per la successiva divisione cellulare.



Divisione Cellulare: MITOSI

Interfase: G1 to S transition



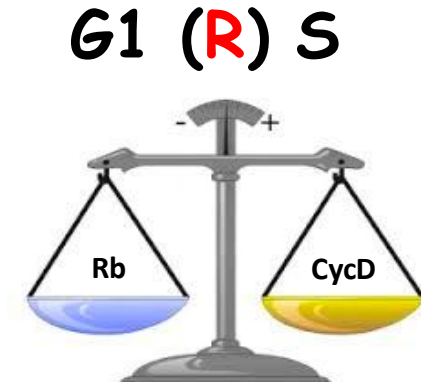
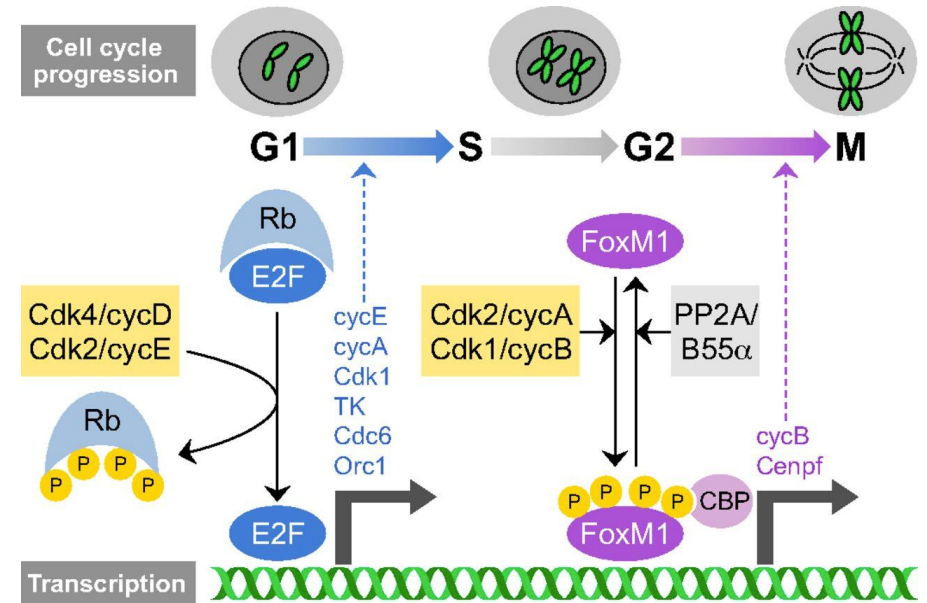
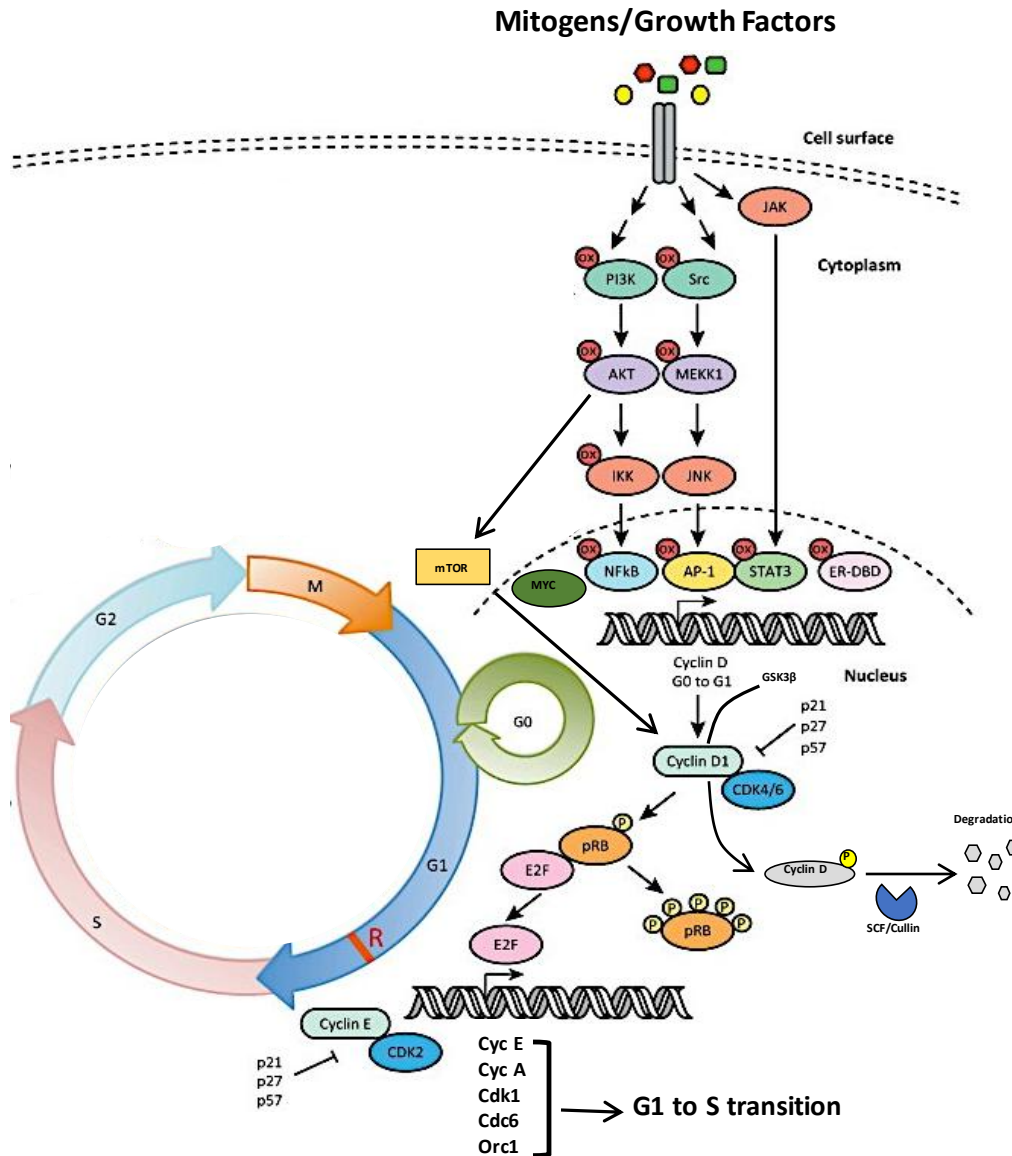
THE RESTRICTION POINT (*R*)

The cell will only pass the **G₁ checkpoint** if size, amount of building molecules and energy are enough.

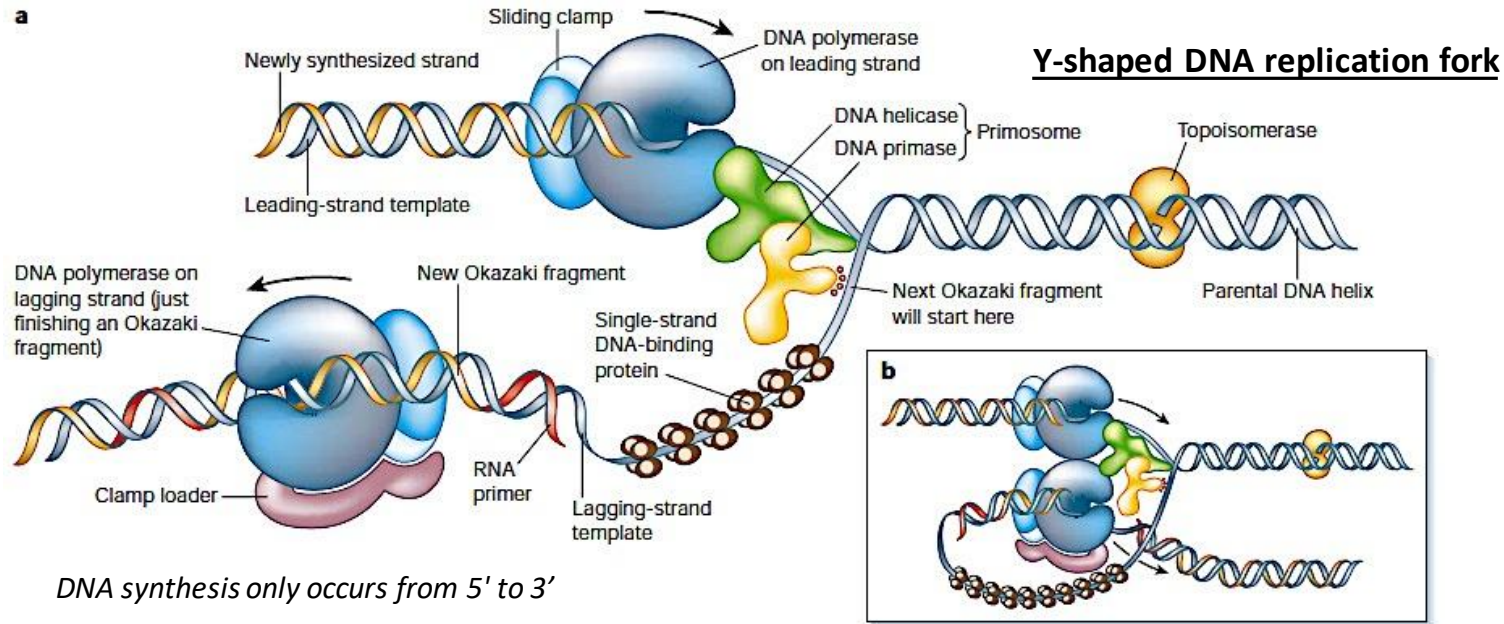
If a cell meets the requirements to pass through the G₁ checkpoint, then it enters the S phase and begins DNA replication.



The G1 → S Restriction Checkpoint



The S Phase: Genome Replication

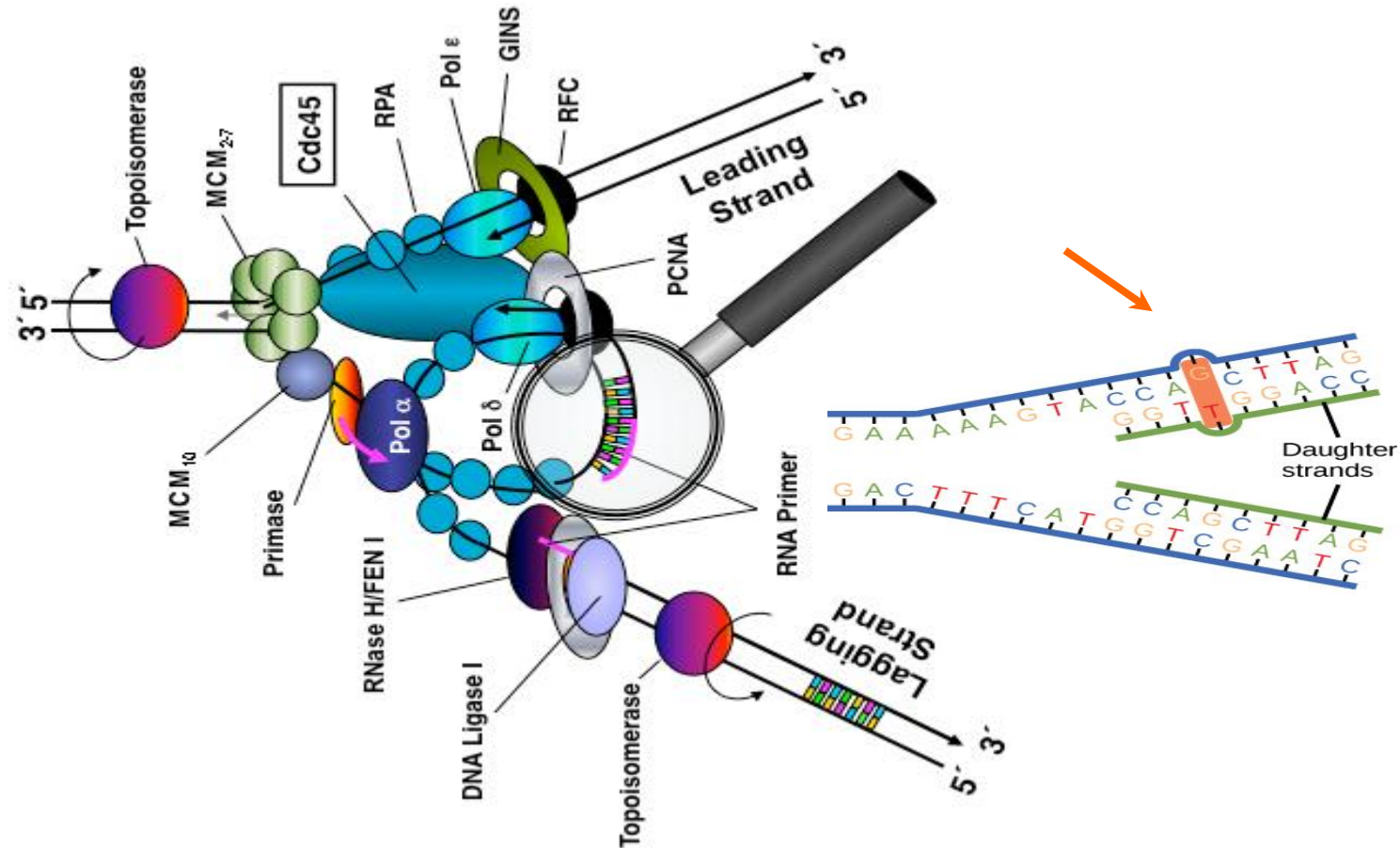


- **DNA topoisomerase enzymes facilitate DNA helix unwinding** while a DNA helicase, powered by ATP hydrolysis, propels itself rapidly along one of the template DNA strands forcing open the DNA helix ahead of the replication fork. The helicase exposes the bases of the DNA helix for the leading-strand polymerase to copy.
- Two DNA polymerase molecules are active at the fork at any one time. One moves continuously to produce the new daughter DNA molecule on the **leading strand**, whereas the other produces a long series of short Okazaki DNA fragments on the **lagging strand**. In addition to the template, DNA polymerases need a pre-existing DNA or RNA chain end (a primer) onto which to add each nucleotide. For this reason, the lagging strand polymerase requires the action of a DNA primase enzyme before it can start each Okazaki fragment. In the folded fork structure shown in the inset, the lagging-strand DNA polymerase remains tied to the leading-strand DNA polymerase. This allows the lagging-strand polymerase to remain at the fork after it finishes the synthesis of each Okazaki fragment.
- Finally, the single-stranded regions of DNA at the fork are covered by multiple copies of a single-strand DNA-binding protein, which hold the DNA template strands open with their bases exposed.
- An RNaseH enzyme will remove the RNA primers from the Okazaki fragments, and a DNA ligase to seal the adjacent Okazaki fragments together to form a continuous DNA strand.

..... just for 3 billion base pairs, roughly 2.5 meters

DNA DAMAGE: Nucleotide Misincorporation

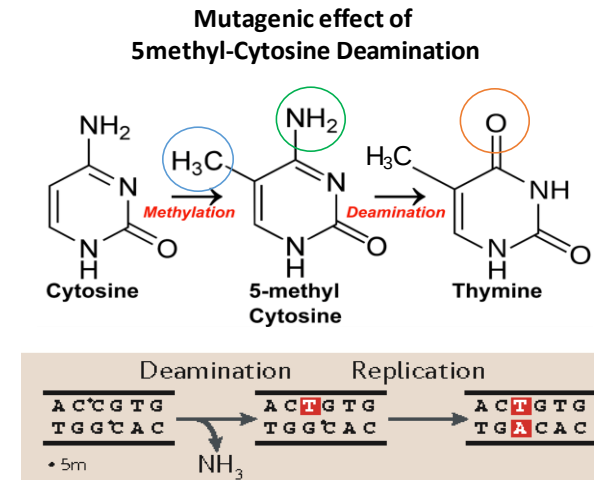
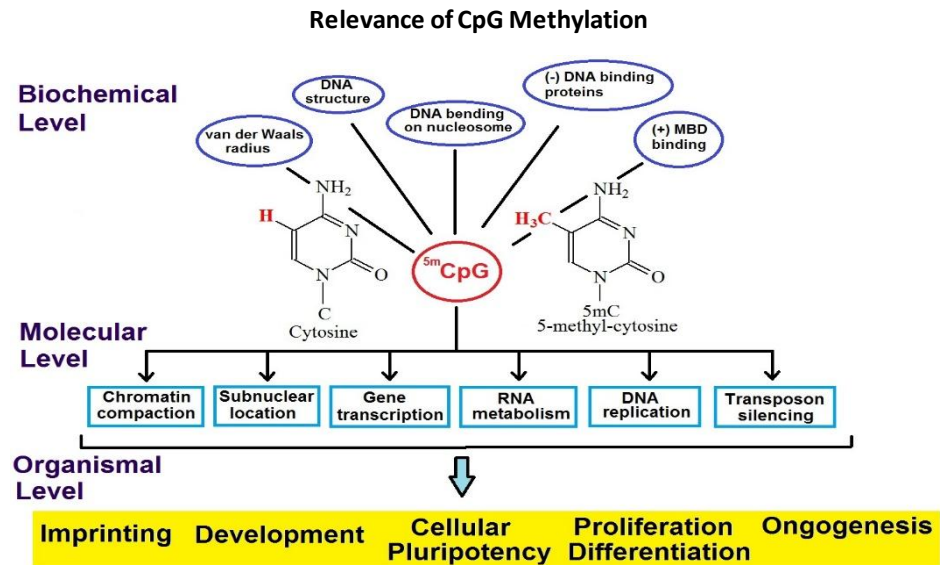
DNA Replication



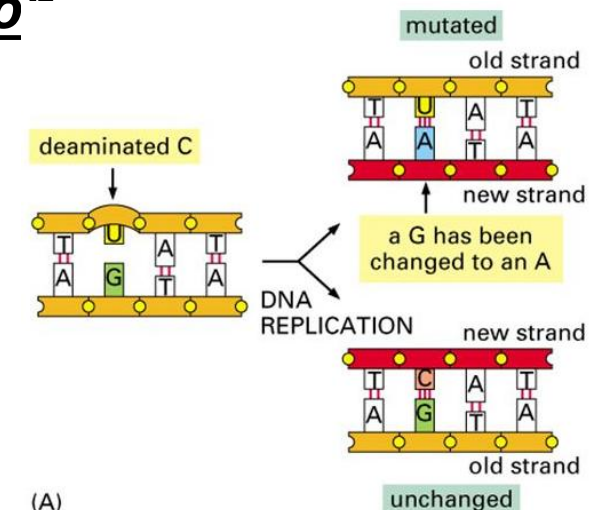
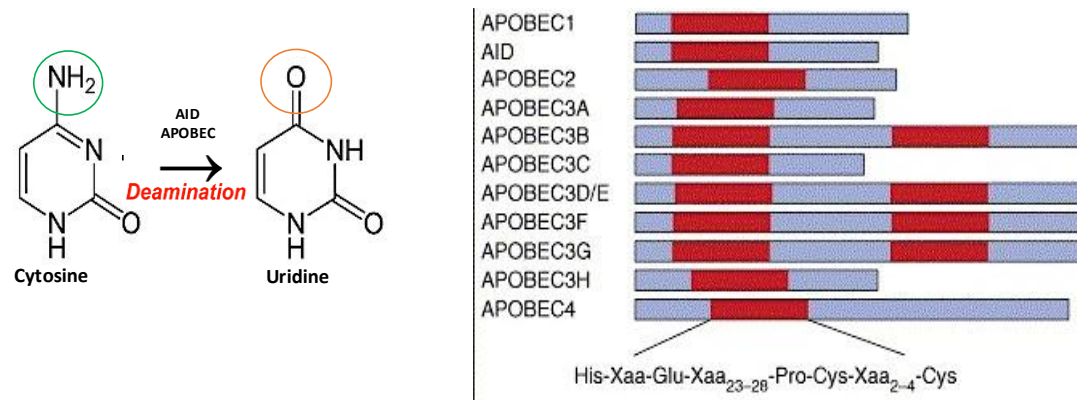
The high-fidelity proofreading **DNA polymerases Pol δ and Pol ε** have an **error rate of 1 in 10⁷.....**
but the **genome length is almost 3x10⁹ bp.**

Furthermore, unbalance of cellular deoxynucleoside triphosphate (dNTP) pool
cause nucleotide misincorporation mainly C:G to T:A or A:T.

Spontaneous 5-methyl Cytosine Deamination

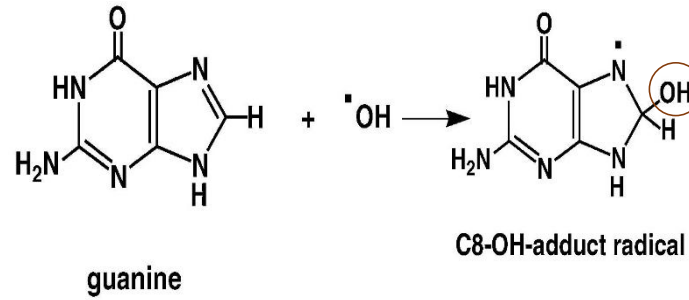
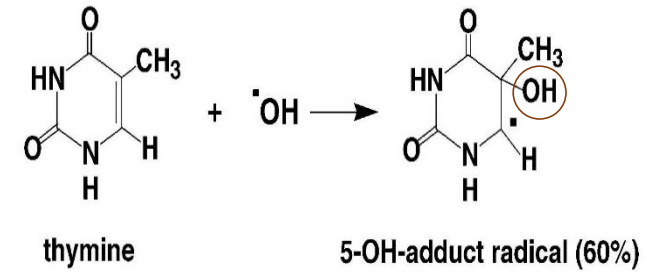
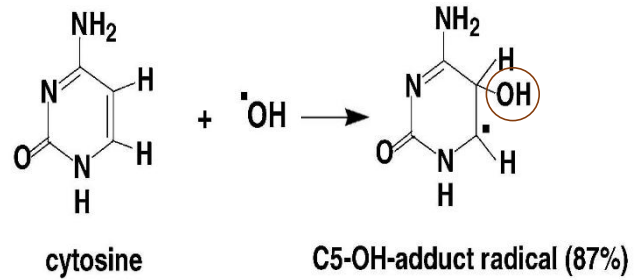


Enzymatic Cytosine Deamination

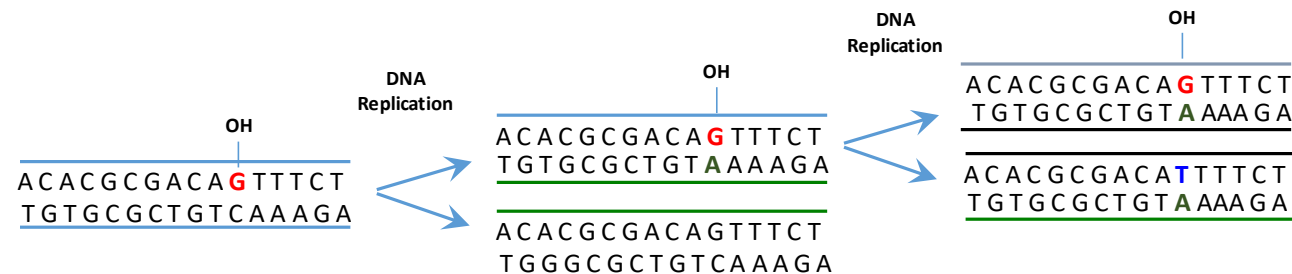


APOBEC enzymes bind single strand DNA with higher affinity than double strand DNA. Notably, this class of enzymes is involved in the mechanism of restriction of retroviruses and mobile retroelements.

Oxidative Damage



For instance, 8-OH-Gua (**G**) results in G → T substitution,

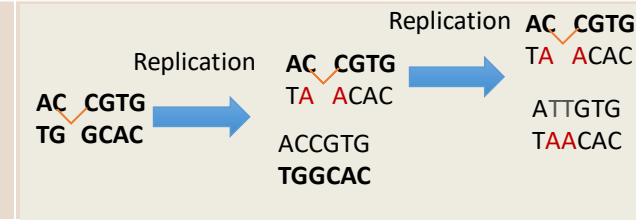


DNA stressors causing Nucleotide Misincorporation

UV radiation



UV radiation on pyrimidines and dipyrimidines



UV radiation causes covalent modifications between neighboring pyrimidine nucleotides. These modifications result in pyrimidine dimers which during DNA replication are misinterpreted by DNA-Pol.

CC:GG to TT:AA substitution are typical of skin cancers.

Tobacco smoking

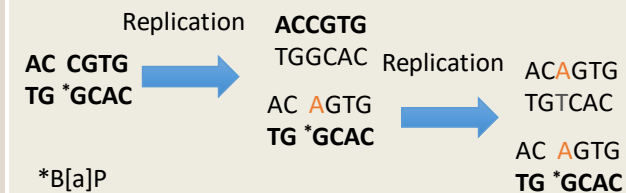


Chemical in Cigarette Smoke

Acetone
Acetic Acid
Ammonia
Arsenic
Benzene
Butane
Cadmium
Carbon Monoxide
Formaldehyde
Hexamine
Lead
Napthalene
Methanol
Nicotine
Tar
Toluene

Benzo[a]pyrene (B[a]P) diol epoxides, a carcinogenic by-product of tobacco smoking, cause G·C→T·A transversions.

Benzo[a]pyrene (B[a]P) adducts on guanine

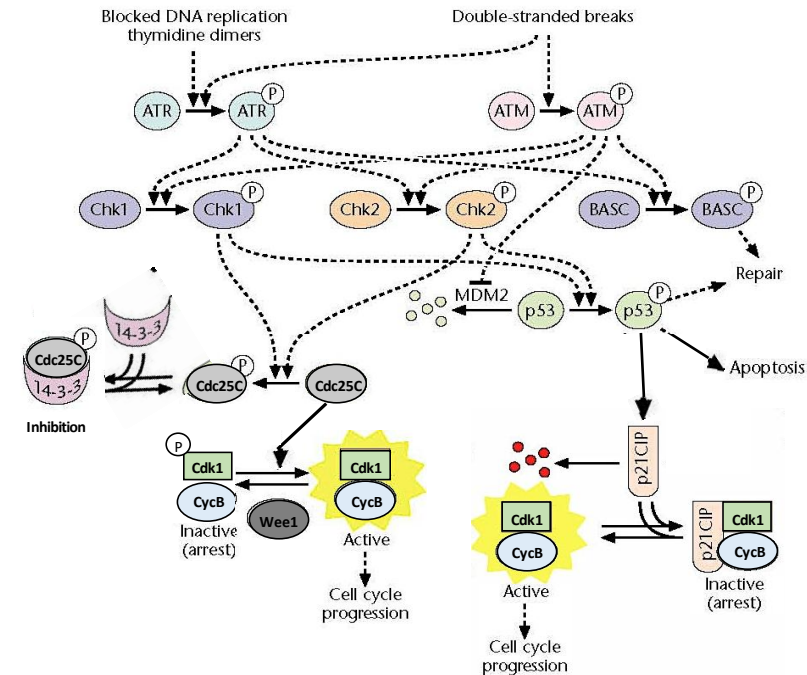
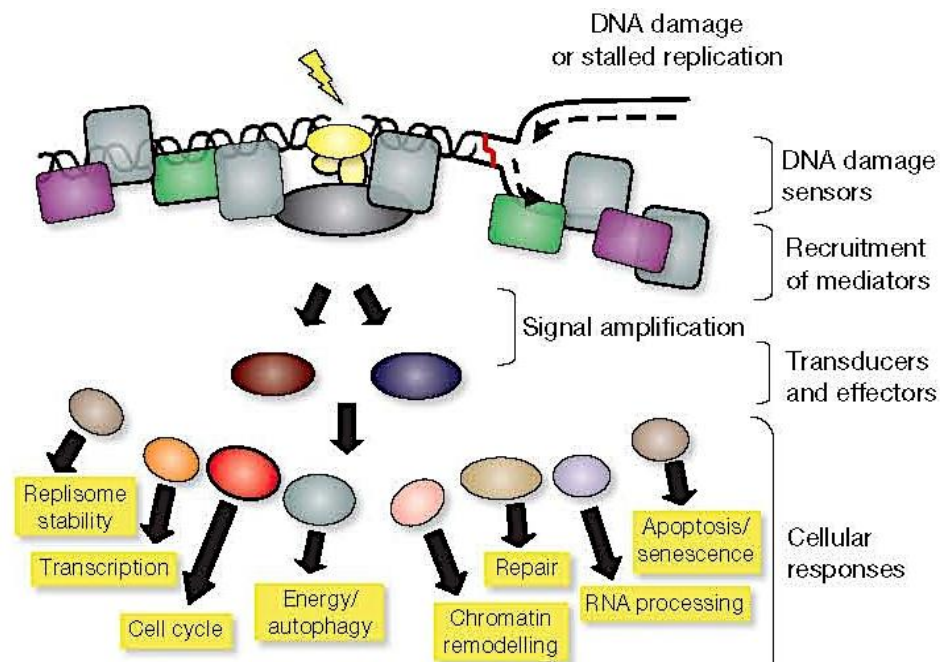


DNA Damage Response

Each of the 10^{13} cells in the human body receives tens of thousands of DNA lesions per day (i.e. strong sunlight can induce ~100.000 lesions per exposed cell per hour).

To cope with constant assault on genomic integrity, cells have evolved the ‘**DNA damage response (DDR)**’:

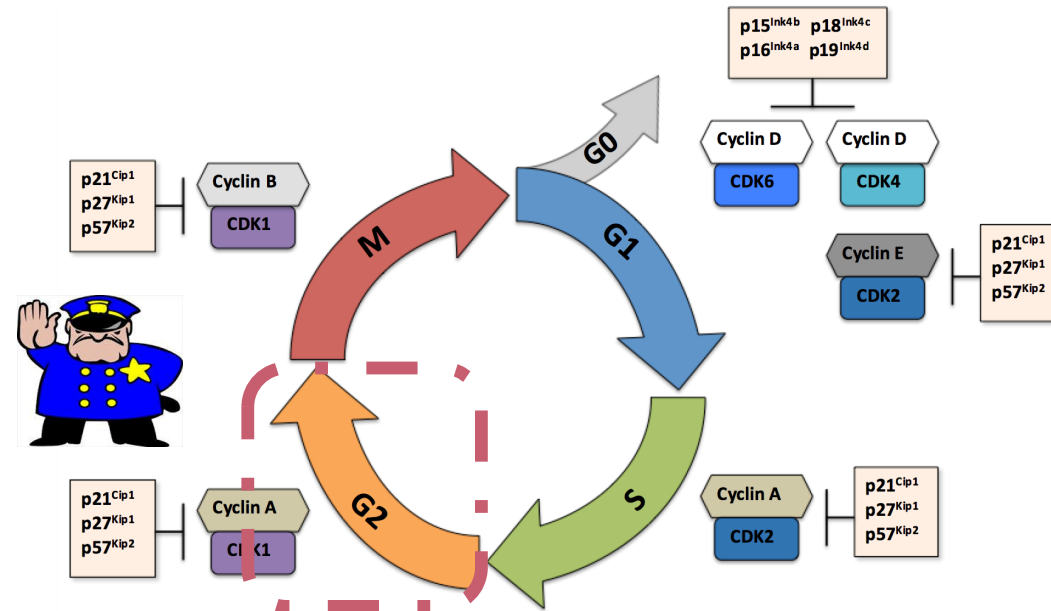
1. DDR detects DNA lesions
2. Initiates checkpoints that arrest the ongoing cell cycle
3. Activates dedicated DNA repair pathways
4. If the amount of DNA damage exceeds the repair capacity, DDR will clear damaged cells through senescence or apoptosis.



The G2 Phase: The last check before Mitosis

Before undergoing chromosome condensation and nuclear division, cells have to be sure that their DNA is fully replicated and undamaged (DSBs).

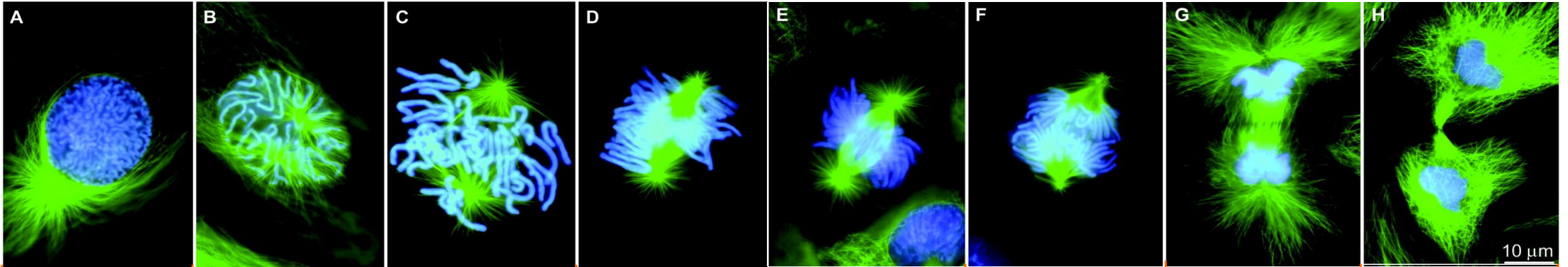
The most important role of the G2 checkpoint is to ensure that all of the chromosomes have been accurately replicated without mistakes or damage. If the checkpoint mechanisms detect problems with the DNA, the cell cycle is stopped, and the cell attempts to either complete DNA replication or repair the damaged DNA. If the DNA has been correctly replicated, cyclin dependent kinases (CDKs) signal the beginning of mitotic cell division. If damage is extensive, cell can activate the apoptotic program.



Mechanisms of DNA Repair

- Base Excision Repair (BER)
- Nucleotide Excision Repair (NER)
- Homologous Recombination (HR)
- Non-Homologous End-Joining (NHEJ)
- Micro-homology Mediate End Joining (MMEJ)

M phase: Cell Division



Prophase

During prophase, which occurs after G2 interphase, the cell prepares to divide by **tightly condensing its chromosomes** and **initiating mitotic spindle formation**. At the onset of prophase, chromatin fibers condense into discrete chromosomes that are typically visible at high magnification through a light microscope. Gene transcription ceases during prophase and does not resume until late anaphase to early G1 phase. The nucleolus also disappears during early prophase.

Prometaphase

At the beginning of prometaphase, phosphorylation of nuclear lamins causes the **nuclear envelope to disintegrate into small membrane vesicles**. As this happens, microtubules invade the nuclear space. A number of **polar microtubules find and interact with corresponding polar microtubules** from the opposite centrosome to form the mitotic spindle. In late prometaphase, microtubules begin to **search for and attach to chromosomal kinetochores**. A kinetochore is a proteinaceous microtubule-binding structure that forms on the chromosomal centromere during late prophase.

Metaphase

After the microtubules have located and attached to the kinetochores in prometaphase, the **two centrosomes begin pulling the chromosomes towards opposite ends of the cell**. The resulting tension causes the chromosomes to **align along the metaphase plate or equatorial plane**, an imaginary line that is centrally located between the two centrosomes (at approximately the midline of the cell). To ensure equitable distribution of chromosomes at the end of mitosis, the metaphase checkpoint guarantees that kinetochores are properly attached to the mitotic spindle and that the chromosomes are aligned along the metaphase plate. If the cell successfully passes through the metaphase checkpoint, it proceeds to anaphase.

Anaphase

During anaphase A, the **cohesins that bind sister chromatids together are cleaved**. **Shortening of the kinetochore microtubules pulls the newly separated daughter chromosomes** to opposite ends of the cell. During anaphase B, polar microtubules push against each other, causing the cell to elongate.

Telophase

At telophase, a **new nuclear envelope forms** using the membrane vesicles of the parent cell's old nuclear envelope. The new envelope forms around each set of separated daughter chromosomes (though the membrane does not enclose the centrosomes) and the nucleolus reappears. **Both sets of chromosomes, now surrounded by new nuclear membrane, begin to "relax" or decondense.**

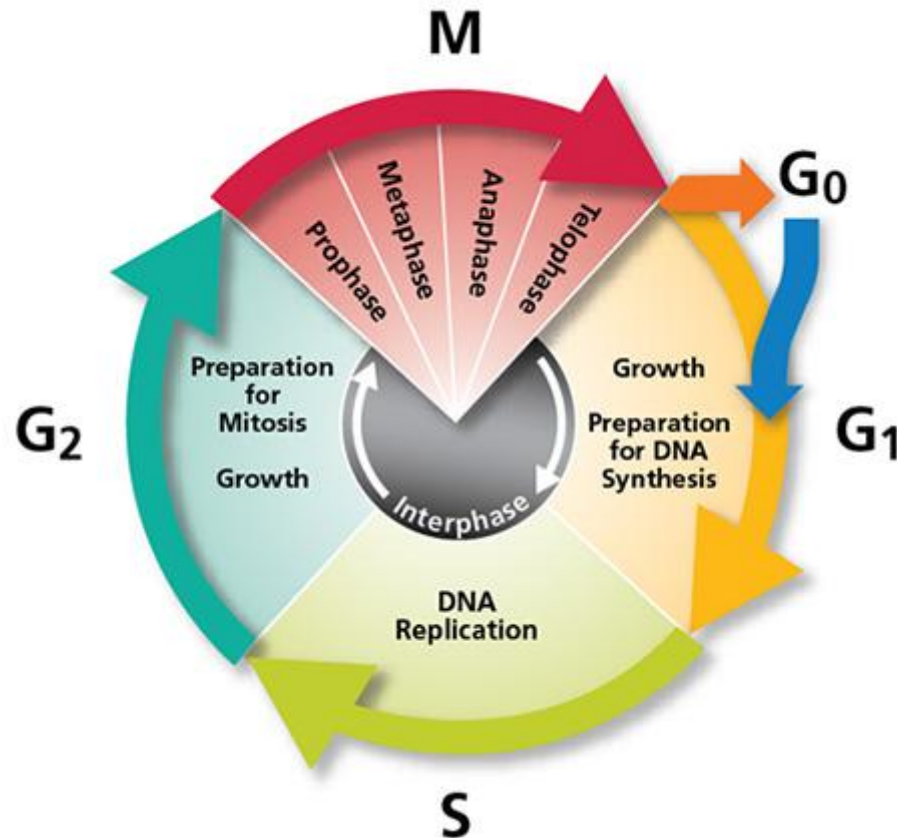
Cytokinesis

Cytokinesis is not a phase of mitosis but rather a separate process, necessary for completing **cell division**. In animal cells, a cleavage furrow (pinch) containing a contractile ring develops where the metaphase plate used to be, pinching off the separated nuclei.

M Checkpoint: Controlling chromosomes-spindle attachment



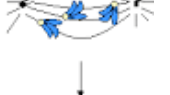
The M checkpoint is also known as the spindle checkpoint because it determines whether all the sister chromatids are correctly attached to the spindle microtubules. Because the separation of the sister chromatids during anaphase is an irreversible step, the cycle will not proceed until the kinetochores of each pair of sister chromatids are firmly anchored to at least two spindle fibers arising from opposite poles of the cell.





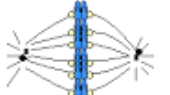
Prometaphase

Chromosomes associated to mitotic spindle



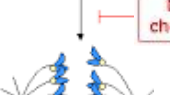
Metaphase

Chromosomes aligned in the metaphase plate

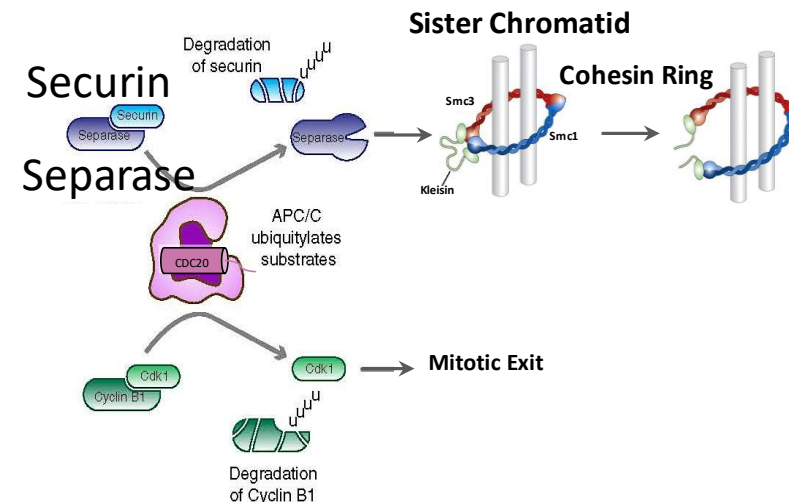
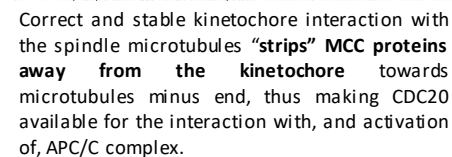
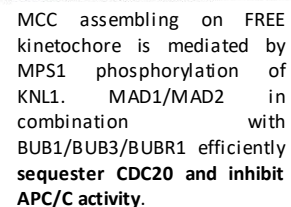
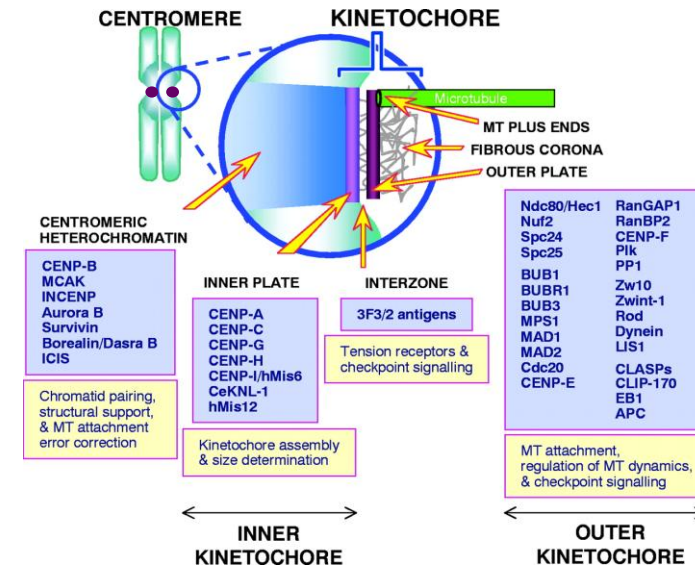


Anaphase

Sister chromatid separation



Mitotic checkpoint



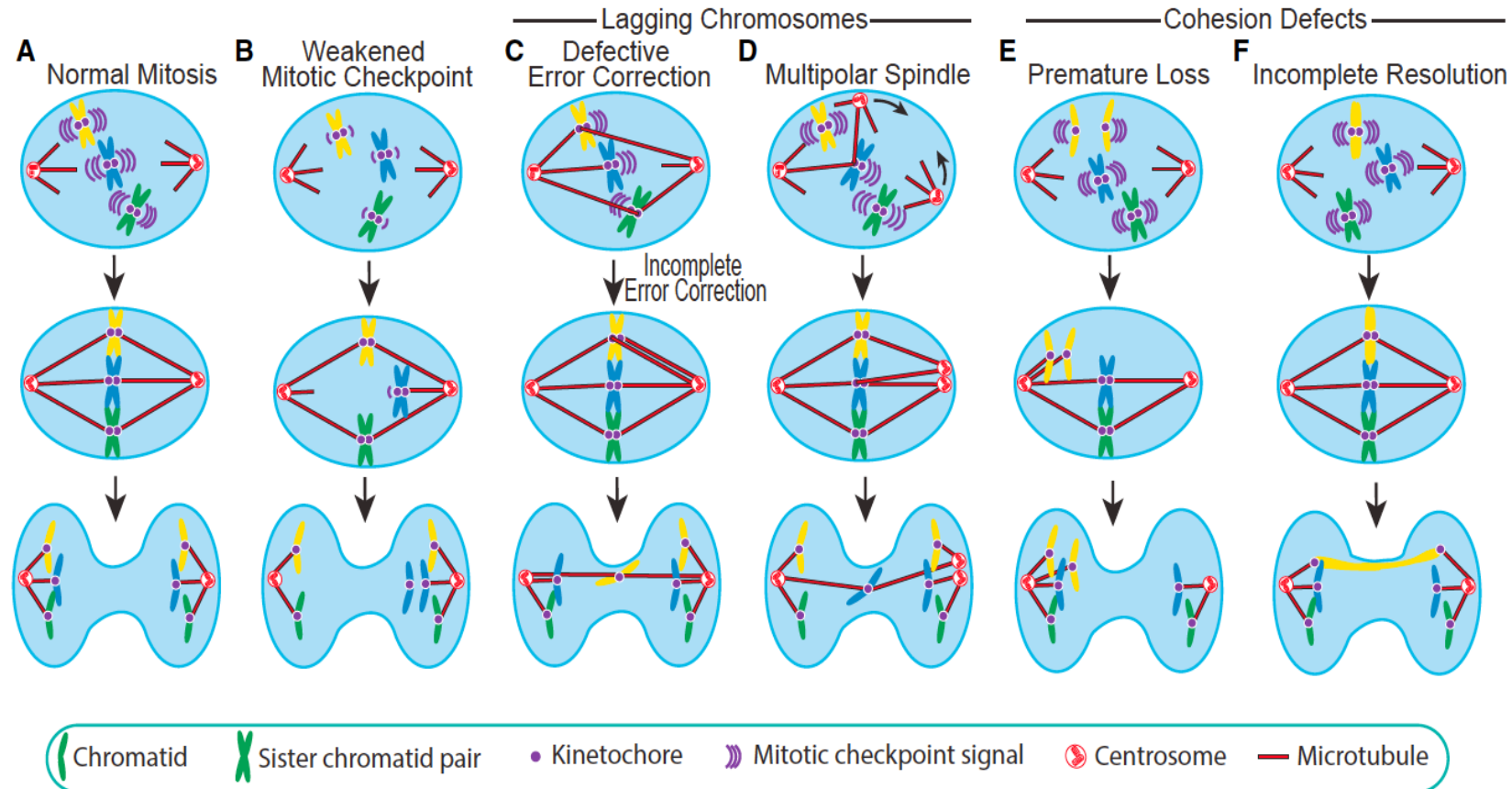
Cell Cycle: Chromosomal Instability-CIN

CellPress

Developmental Cell
Review

Living in CIN: Mitotic Infidelity and Its Consequences for Tumor Promotion and Suppression

Laura C. Funk,¹ Lauren M. Zasadil,¹ and Beth A. Weaver^{2,*}



Divisione Cellulare: La MEIOSI

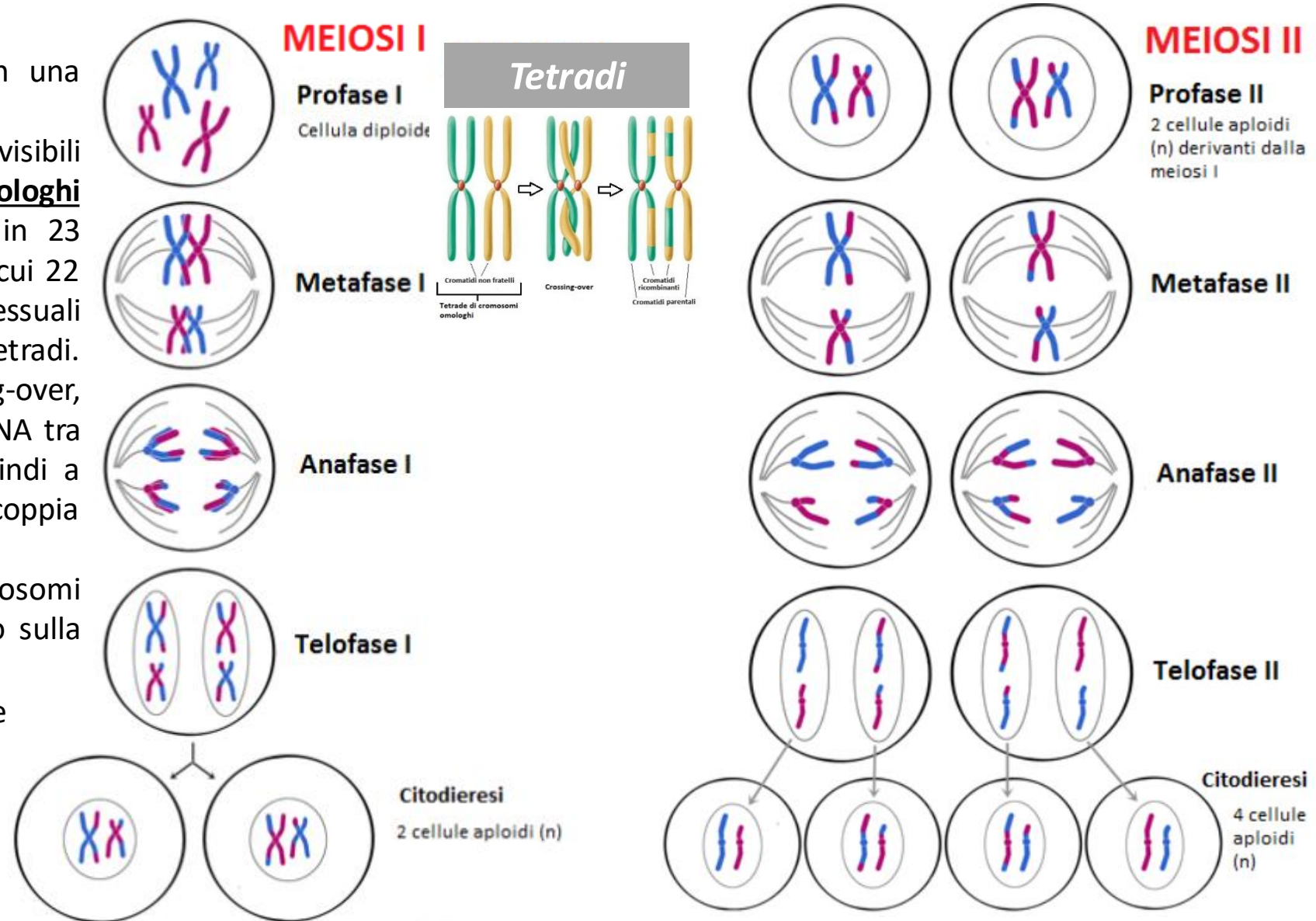
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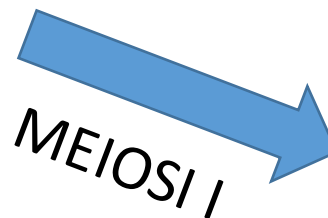
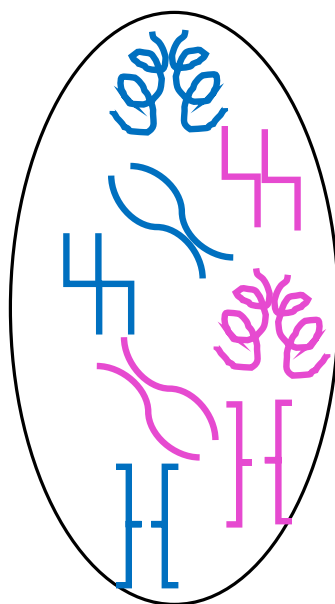
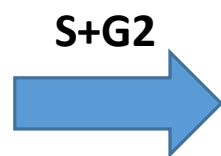
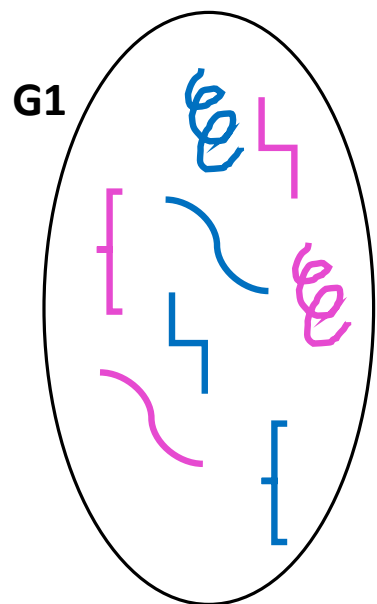
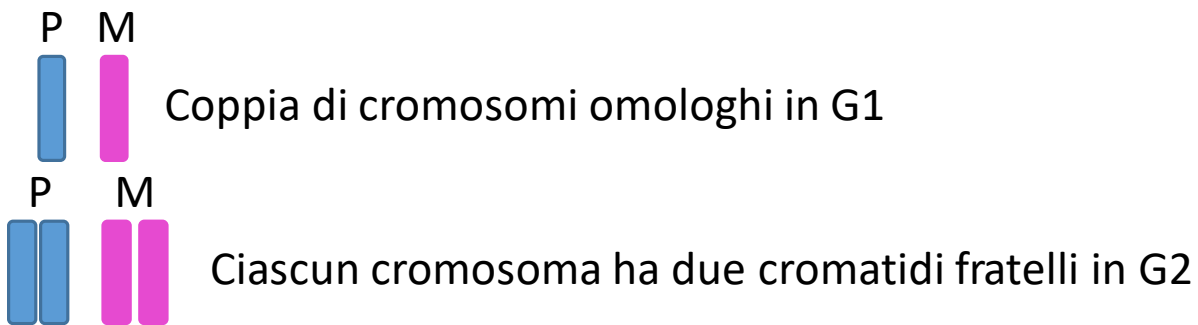
La cellula duplica il proprio DNA come in una classica Mitosi.

In **Profase I** (*prima*) i cromosomi diventano visibili ed l'appaiamento dei cromosomi omologhi ((essere umano: 46 cromosomi distribuiti in 23 coppie (1M+1P) di cromosomi omologhi di cui 22 autosomici e una coppia di cromosomi sessuali (eterosomi)) consente la formazione delle tetradi. Arrivati a questo punto avviene in crossing-over, cioè uno scambio di porzioni omologhe di DNA tra due cromatidi non fratelli, appartenenti quindi a due diversi cromosomi facenti parte di una coppia di omologhi.

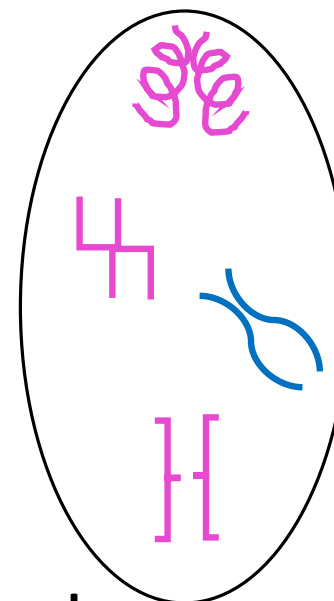
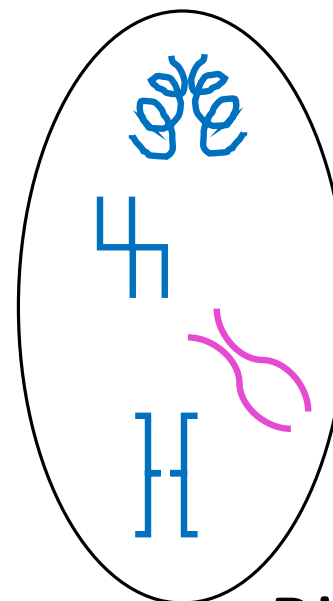
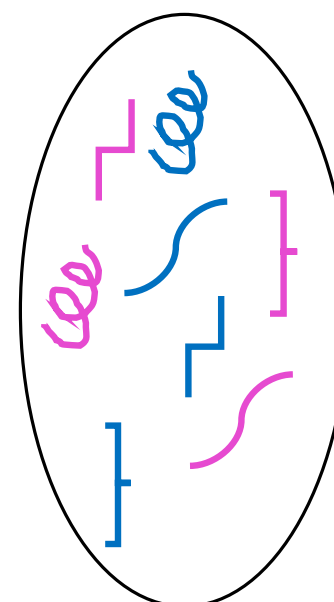
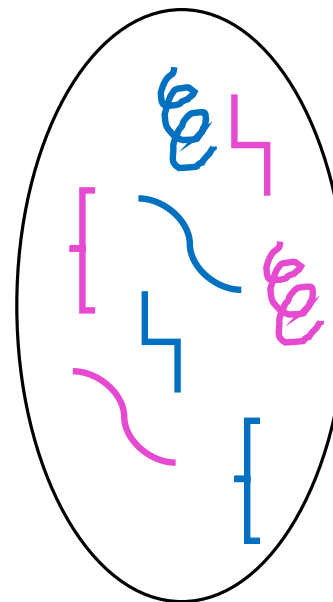
in **Metafase I** grazie all'interazione dei cromosomi con il fuso mitotico le tetradi si dispongono sulla piastra mitotica.

Nel corso dell'**Anafase I** avviene la separazione dei cromosomi omologhi, che migrano ai *poli* opposti della cellula madre. La distribuzione di ciascuna coppia di omologhi ai due poli è totalmente casuale.





Conservativa



Riduzionale